TEST PLAN DOCUMENT

SPRINT 7

1. Test Plan Objective

Objective:

The objective of this Test Plan is to verify that the Employee Management System (EMS) meets all functional and non-functional requirements defined in Sprint 6 and is ready for final delivery. The plan ensures that all implemented features work correctly after code cleanup, refactoring, and integration, and that the system is usable, stable, and aligned with Product Owner expectations.

2. Scope of Testing

Include:

In Scope

- Login & Authentication
- Dashboard statistics
- Employee Management (CRUD)
- Department Management (CRUD)
- Leave Management
- Salary Management
- Localization (English, Spanish, Persian, Arabic)
- Responsiveness
- Error handling
- Security basics (password validation, restricted access)

Out of Scope

- Performance benchmarking
- Load testing
- External integrations (none in your project)

3. Test Resources

List the people and tools used.

Human Resources

- Test Engineer 1: Amir Noori
- Test Engineer 2: Jesper Holmström
- Test Engineer 3: Mehdi Zaidane
- Test Engineer 3: Osama Aamer

Tools

- IntelliJ IDEA
- Browser (Chrome / Edge)
- Jenkins CI/CD
- SonarQube (static code analysis)
- Docker (deployment)
- GitHub (source control)
- Trello (user stories & sprint tracking)
- Excel / Google Sheets (bug tracking & UAT results)

4. Test Environment

Document the environment where testing is performed:

Software Environment

- Backend: Java
- Frontend: JavaFx
- Database: MariaDB / MySQL
- Build Tool: Maven
- Deployment: Docker container
- CI/CD: Jenkins pipeline
- Static Analysis: SonarQube
- Browser: Chrome (latest)

Test Data

- Admin test account

- Sample employees
- Sample departments
- Sample leave records
- Sample salary entries

5. Test Tasks (Functional & Non-Functional)

This is the most important part.

Functional Test Tasks

1. Execute unit tests for all modules
2. Perform integration testing (Dashboard → Employees → Departments → Leave → Salary)
3. Validate CRUD operations
4. Validate form validation and error messages
5. Validate localization switching
6. Validate navigation and responsiveness
7. Validate login/logout behavior
8. Validate salary calculations and leave duration calculations
9. Execute UAT test cases (10 test cases from Sprint 6)

Non-Functional Test Tasks

1. **Heuristic Evaluation** using Nielsen's 10 heuristics
2. **User Acceptance Testing (UAT)**
3. **Performance checks** (page load < 2 seconds)
4. **Security checks** (password rules, restricted pages)
5. **Responsiveness checks** (forms, tables, dashboard)
6. **Code quality analysis** using SonarQube
7. **Documentation of technical changes** after testing

6. Bug Tracking Process

Explain how bugs will be tracked:

- All bugs will be recorded in a shared Excel/Google Sheet
- Each bug will include:
    - Bug ID
    - Description
    - Severity (Low/Medium/High)
    - Module
    - Steps to reproduce
    - Status (Open/Fixed/Retest)
    - Assigned to
    - Date found

This table will be included in the Sprint 7 report.

7. Acceptance Criteria

State that the system will be accepted if:

- All **critical** and **major** bugs are fixed
- All UAT test cases pass
- SonarQube grades are **A or B**
- No broken UI or missing translations
- All core features function correctly
- Localization works for all 4 languages
- System is stable and deployable via Jenkins + Docker

8. Risks & Mitigation

Include:

Potential Risks

- Incomplete localization
- UI responsiveness issues
- Missing error messages

- Jenkins pipeline failures
- SonarQube failing grades

Mitigation

- Daily testing
- Immediate bug fixing
- Re-running Jenkins pipeline
- Code cleanup and refactoring
- Peer review of UI changes

9. Test Schedule

A simple timeline:

- Day 1–2: Unit testing + integration testing
- Day 3: UAT execution by all team members
- Day 4: Heuristic evaluation
- Day 5: Bug fixing + retesting
- Day 6: Jenkins + SonarQube + Docker deployment
- Day 7: Final report preparation

10. Test Deliverables

You will produce:

- Test Plan
- Unit test results
- Bug tracking table
- Heuristic Evaluation Report
- UAT Results Report
- Jenkins build screenshots
- SonarQube report
- Updated architecture documentation
- Sprint 7 final report